Date: 02 July 2026	**Team ID:** SB-7429
Project Name: Credit Card Approval Prediction using Machine Learning	**Maximum Marks:** 5 Marks

PROBLEM-SOLUTION FIT TEMPLATE

Problem-Solution Fit Canvas

Fill out the canvas in the respective boxes below to ensure your proposed solution addresses the pains, root causes, and limitations of the target user segments.

1. CUSTOMER SEGMENTS **- Commercial banks** **- Digital fintech lenders** **- Credit officers** **- Credit card applicants**	**2. PROBLEMS / PAINS** **- High underwriting latency (days)** **- High cost per application** **- Human errors & subjective bias** **- Default risk (NPAs)**	**3. TRIGGERS TO ACT** **- High customer abandonment rate** **- Escalating NPA default rates** **- Auditor pressure for compliance**
4. EMOTIONS (Before/After) - Before: Anxious, frustrated, overwhelmed - After: Relieved, confident, satisfied	5. AVAILABLE SOLUTIONS - Manual underwriters (slow) - Heuristic scorecards (rigid rules, miss non-linear risk patterns)	6. CUSTOMER LIMITATIONS - Constrained local servers - Computer security policies blocking dynamic XGBoost DLLs
7. BEHAVIOR & INTENSITY - Risk officers cross-check files manually (high intensity, low throughput)	8. CHANNELS OF BEHAVIOR - Online: Web underwriting portals - Offline: CSV training logs	9. PROBLEM ROOT CAUSE - Linear models fail to capture multi-dimensional correlations

10. YOUR PROPOSED SOLUTION (SL) **- A hybrid underwriting engine combining Layer 1 Policy Override Rules and a Layer 2 SMOTE-balanced, calibrated Random Forest Classifier, deployed as a secure dark-theme web dashboard on Render.**